UT-26 MAPLE
Multicopter Autonomous Package Logistics Enabler

SAE AERO DESIGN WEST

ADVANCED CLASS DESIGN REPORT

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University *of* Toronto Aerospace Team
Team 206

Contributors:
Sencer Aksu, Kai Chen, Jensen Langlois, Michael Law, Kevin Wang
Yume Yamamoto, Frank Yan, Aryan Athavale, Satvik Garg,
Arthur Hwang, Vitto Janapin, Ethan Liu, Leon Liu, Jason Miao,
Alex Shen, Barnett Wan, Tszhei Wong, Kevin Xing, George Xue

APPENDIX A - STATEMENT OF COMPLIANCE

Certification of Qualification

Team Name	U of T Aerospace Team	Team Number	206
School	Univ of Toronto		
Faculty Advisor	Peter Grant		
Faculty Advisor's Email	peter.grant@utoronto.ca		

Statement of Compliance

As faculty Adviser:

RS ______ (Initial) I certify that the registered team members are enrolled in collegiate courses.

RS ______ (Initial) I certify that this team has designed and constructed the radio-controlled aircraft in the past nine (9) months with the intention to use this aircraft in the 2026 SAE Aero Design competition, without direct assistance from professional engineers, R/C model experts, and/or related professionals.

RS ______ (Initial) I certify that this year's Design Report has original content written by members of this year's team.

RS ______ (Initial) I certify that all reused content has been properly referenced and is in compliance with the University's plagiarism and reuse policies.

RS ____ (Initial) I certify that the team shall use the Requirements Check & Safety and Airworthiness Inspection checklists to inspect their aircraft before arrival at Technical Inspection and that the team shall submit the completed checklists, signed by the Faculty Advisor or Team Captain, to the inspectors before Technical Inspection begins.

______________________	February 25th, 2026
Signature of Faculty Advisor	Date
______________________	February 25th, 2026
Signature of Team Captain	Date

Note: A copy of this statement needs to be included in your Design Report as page 2 (Reference Section 4.3)

Executive Summary

SYSTEM OVERVIEW

Designed by the University of Toronto Aerospace Team, UT-26 MAPLE is a multicopter fixed-wing V/STOL aircraft intended for autonomous package delivery and capture missions. The aircraft features a lifting-body tri-copter airframe with a tricycle landing gear configuration, primarily fabricated of balsa, with carbon fiber reinforced plastics for primary load-bearing structures.

COMPETITION PROJECTIONS

MAPLE is projected to carry a payload of 0.22 lb and execute a fully autonomous flight, consisting of a conventional takeoff, transition to vertical flight, payload delivery and capture, and a conventional return to base landing. MAPLE is expected to attain a cumulative flight score of 35.82.

DISCRIMINATORS

The majority of the aircraft's design was performed using in-house tools, specially adapted over the past five years for small unmanned aerial vehicle designs. This includes a multidisciplinary optimization of the wing planform, and a custom tail-sizing algorithm to determine compatible tail sizes and center-of-gravity ranges for desirable stability and control characteristics. Additionally, three prototype aircraft were built prior to MAPLE to rapidly test and mature newly upgraded or developed software systems.

MAPLE's software systems were developed through comprehensive validation in both simulation and real-life environments, rigorous evaluation of a wide range of approaches, and a requirements-driven design process to maximize competition scoring potential while mitigating technical risks. Its fully autonomous target localization and precision landing system integrates state-of-the-art computer vision techniques and unique control strategies to deliver a robust, precise, and reliable performance.

UT-26 as-built.

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Nomenclature

δ	Control surface deflection
η	Propulsion efficiency
η_m	Motor efficiency
η_p	Propellor efficiency
b	Wing span
c	Wing chord
$c/4$	Wing quarter chord
C_L	Lift coefficient
C_M	Moment coefficient
C_{D0}	Profile drag coefficient
$C_{L,\max}$	Maximum lift coefficient
m	Mass
m_{OEW}	Operating empty weight
n	Load factor
S	Wing area
s_{runway}	Runway length
S_h/S_w	Tail-to-wing area ratio
V	Velocity
V_{stall}	Stall speed
x_{cg}	Lengthwise centre of gravity location
y	Spanwise axis
AR	Aspect ratio
BEC	Battery eliminator circuit
CAD/CAE	Computer-aided design/engineering
CFRP	Carbon fiber-reinforced plastic
CG	Centre of gravity
COTS	Commercial off-the-shelf
CV	Computer vision
DAS	Data acquisition system
DLZ	Designated landing zone
ESC	Electronic speed controller
FCU	Flight control unit
FEA	Finite element analysis
FMEA	Failure modes abd effects analysis
FOV	Field-of-view
GCS	Ground control station
IMU	Inertial measurement unit
MAC, $\bar{c}$	Mean aerodynamic chord
MDO	Multidisciplinary design optimization
MTOW	Maximum takeoff weight
NP	Neutral point
OML	Outer mold line
PCB	Printed circuit board
PID	Proportional-integral-derivative
Re	Reynolds Number
ROS	Robot Operating System
RTK	Real-time kinematic
T/W	Thrust-to-weight ratio
TRL	Technology readiness level
UTAT	University of Toronto Aerospace Team
V/STOL	Vertical and/or short takeoff and landing

1 Strategy

1.1 Design Requirements

The Final Flight Score is the sum of the top three competition flight scores, calculated using the following equation. Each flight comprises four mission segments: conventional takeoff, payload delivery, payload capture, and conventional landing/return to base. Figure 1.1 illustrates the mission profile for each flight at the Thunderbirds RC Club in Fort Worth, TX. Select mission requirements, which guided the design and testing efforts, are listed in Table 1.1.

$$\text{Flight score} = \sum_{x}^{4} S_x + TB$$

$$\text{where } S_x = S_{\text{mission segment}} = 1 + (MSM \times W_{\text{payload}})$$

$$TB = \text{Time bonus} = \begin{cases} 0, & \text{if } 180 \leq T_{\text{mission}} < 240 \text{ s} \\ 1, & \text{if } 120 \leq T_{\text{mission}} < 180 \text{ s} \\ 2, & \text{if } 0 \leq T_{\text{mission}} < 120 \text{ s} \end{cases}$$

MSM = Mission Segment Multiplier =

Mission segment	Autonomous multiplier	Manual multiplier
Conventional takeoff	2	1
Payload release	4	1
Payload delivery	8	1
Payload capture	14	2
Return to base	3	1

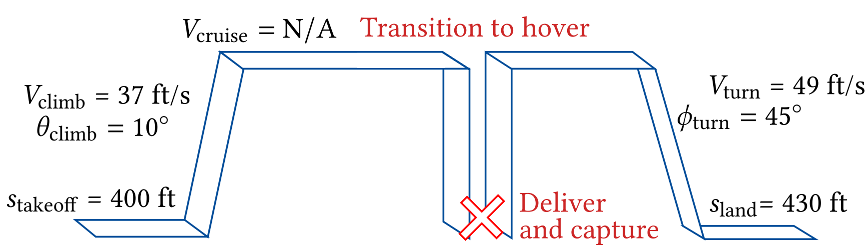


Figure 1.1 Aircraft mission profile.

Table 1.1 System requirements

Mission conditions		
MC-01	Max. takeoff weight	3.5 lb
MC-02	Flight time	4 min
Performance		
PF-01	Takeoff distance	400 ft
PF-02	Climb gradient	10°
PF-03	Bank angle	45°
PF-04	Landing distance	430 ft
PF-05	Stall speed	32 ft/s
Aerodynamics		
AD-01	Max. wing span	10 ft
AD-02	Wing tip stalls after root	
Structures		
ST-01a	Safety factor to yield (major structures)	2.0
ST-01b	(minor structures)	1.5
ST-02	Gust speed	15 ft/s
ST-03	Max. speed	$1.4V_{\text{maneuver}}$
ST-04a	Max. wing tip deflection (torsion)	-2°
ST-04b	Max. wing tip deflection (bending)	$0.1b_{\text{half}}$
ST-05	Min. propeller clearance	2 in
ST-06	Sink gradient @ landing	15°
Stability and control		
SC-01	Tail stalls after wing	
SC-02	Target static margin	10%
SC-03	Min. static margin	5%
SC-04	Max. static margin	15%
SC-05	Min. spiral doubling time at cruise	8 s
SC-06	Aileron: Bank to 60° at $1.13V_{\text{stall}}$	1 s
SC-07	Elevator: Runway pitch to $\alpha = 15°$	0.5 s
SC-08	Elevator: Trim deflection margin	5°
SC-09	Rudder: Maintain directional control with single forward motor loss	

Misc. airframe		
MA-01	Max. dimension	40 in
MA-02	Max. assembly time	10 min
MA-03	Max. gap between adjoining subassemblies	1/32 in
MA-04	Flyable in light precipitation	
Propulsion		
PR-01	Max. motors	3
PR-02	Static T/W	1.3
PR-03	Max. battery capacity (4-cell LiPo)	3000 mAh
PR-04	Min. propeller to arming plug distance	9 in
DAS		
DA-01	Communication range	984 ft
DA-02	Range for DLZ detection	[0.33 ft, 9.84 ft]
DA-03	Range for payload detection	[0.164 ft, 4.92 ft]
DA-04	DLZ detection accuracy @ 9.84 ft	1.64 ft
DA-05	DLZ detection accuracy @ 0.33 ft	0.164 ft
DA-06	Resolution of target	0.39 in / pixel
DA-07	Minimum detection update rate	20 Hz
DA-08	Video latency	300 ms
Controller		
CR-01	Maximum overshoot	7%
CR-02	Min. disk margin ($\sigma = 0$)	$\alpha = 0.5$
CR-03	Hover hold accuracy (x-y plane), no wind	1.64 ft
CR-04	Hover hold accuracy (x-y plane), 9.84 ft/s wind	3.28 ft

1.2 RISK ANALYSIS AND MITIGATION

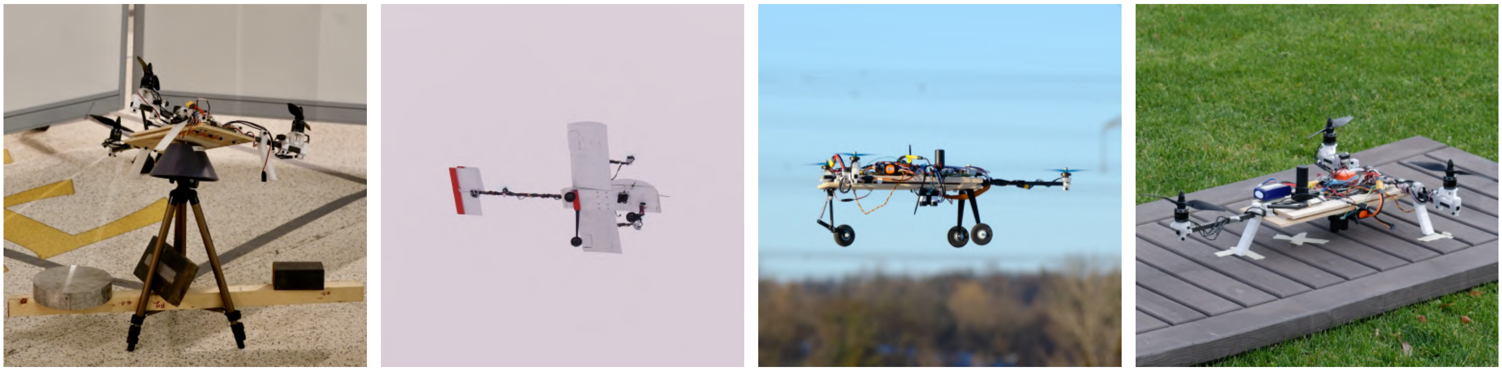
A systems-level FMEA was conducted in the early stages of the design process. The highest technical and operational risk factors are presented in Table 1.2.

Table 1.2 System-level failure modes and effects analysis.

System Function	Failure mode	Effect of failure	Magnitude of effect	Likelihood	Action
Technical risks					
V/STOL aircraft design (structures, propulsion & detailed design)	Delays in design completion	Delayed construction and testing; schedule slip	Medium	High	1. Allocate a 10% schedule buffer for high-risk tasks 2. Consider technical readiness during configuration selection 3. Solicit advisor feedback on critical design decisions 4. Build an iron-bird prototype to identify system integration issues
Fixed-wing controller design, especially during takeoff and landing	Controller performance inadequate or unstable	Aircraft crash during flight tests; delays to downstream system testing	Medium	High	1. Carry out model-based controller design early in the development cycle 2. Deploy and validate the controller on a durable testbed 3. Solicit advisor feedback throughout the controller design and validation process
Rotary-wing controller design	Controller performance inadequate or unstable	Aircraft crash during flight test; delays to downstream testing	High	Low	1. Perform model-based controller design early in the development cycle 2. Deploy and validate the controller on a durable testbed 3. Test and tune controllers on a custom PID tuning stand
Automatic target localization	Failure to land on DLZ or retrieve payload	Reduced points scored in flight rounds	Medium	Medium	1. Deploy to familiar tricopter testbed before the competition spec. aircraft 2. Solicit feedback from advisors throughout the design cycle
Operational risks					
Conduct outdoor flight testing in Toronto	Inclement weather limits flight testing	Delays to flight testing schedule	High	Medium	1. Carry out limited flight operations early in the year 2. Conduct extensive analysis and ground testing
Perform outdoor mission in Fort Worth	Inclement weather compromises flight operations	Poor mission performance	Medium	Medium	1. Design for above-seasonal average winds and light precipitation 2. Conduct test flights in multiple conditions
Competition participation	Aircraft becomes unairworthy before competition	Unable to compete	High	Medium	1. Construct and maintain two backup aircraft 2. Halt flight operations immediately upon detection of critical faults
Manage project finances and budget	Budget overruns or insufficient funding	Unable to fund aircraft construction or attend competition	High	Low	1. Include contingencies in the budget 2. Pursue additional in-kind or monetary sponsorships to supplement team needs
Team availability and expertise	Insufficient technical expertise or personnel	Design disruptions, assembly delays	Low	Medium	1. Cross-train team members 2. Document development processes, designs, and decisions

Two prototypes were modified from previous years or constructed to decouple the development risks identified in Table 1.2 and parallelize the development of software subsystems. Prior to each prototype flight, a PID tuning stand was used to verify calibration and reduce the risk of crashing (Figure 1.2a).

A fixed-wing testbed (dubbed X-26A) was manufactured with a simplified electrical system and no payload mechanism to test fixed-wing controller design (Figure 1.2b) A vertical flight testbed (dubbed X-26B) was designed and built to test propulsion, payload mechanisms, and to serve as an “iron bird” (Figure 1.2c) A flight-proven vertical flight testbed (dubbed X25) was adapted from the previous year, and



(a) PID tuning stand (b) X-26A (c) X-26B (d) X-25

Figure 1.2 Experimental testbeds and test rig utilized for risk reduction.

was used to test automatic payload localization and precision landing capabilities (Figure 1.2d).

These prototypes allowed for multiple aircraft to be developed and flown simultaneously, while decoupling the risks of unproven subsystems.

1.3 PROJECT OVERVIEW

Figure 1.3 shows the high-level design schedule. Project scheduling was determined using an average 5 person-hour/week contribution from each team member. A 15% buffer was included for each scheduled item. The majority of tasks were completed on time, with the exception of the X-26B design and build, which was delayed due to the prioritization of member recruitment and onboarding. The prototype aircraft was eventually completed, and remained flyable during the winter months.

Figure 1.4 shows the distribution of 42 members among 5 subteams. The number of members in each subteams reflects both its assigned level of responsibility and the corresponding workload. The budget for the 2025-26 year is shown in Table 1.3. The funds were raised through university grants, competition earnings, and corporate sponsorships. An 11-member competition roster will attend the event, with part of the costs covered out-of-pocket by the student participants.

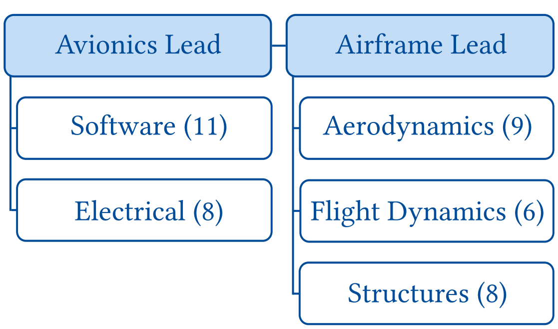


Figure 1.4 UTAT SAE team structure, with the corresponding member count in parentheses.

Table 1.3 2025–26 competition year budget.

Category	Cost ($US)
Construction materials	$ 2 610
Electronics	$ 5 920
Competition travel	$ 11 300
Competition registration	$ 1 650
Misc. administrative costs	$ 360
Total	$ 21 840

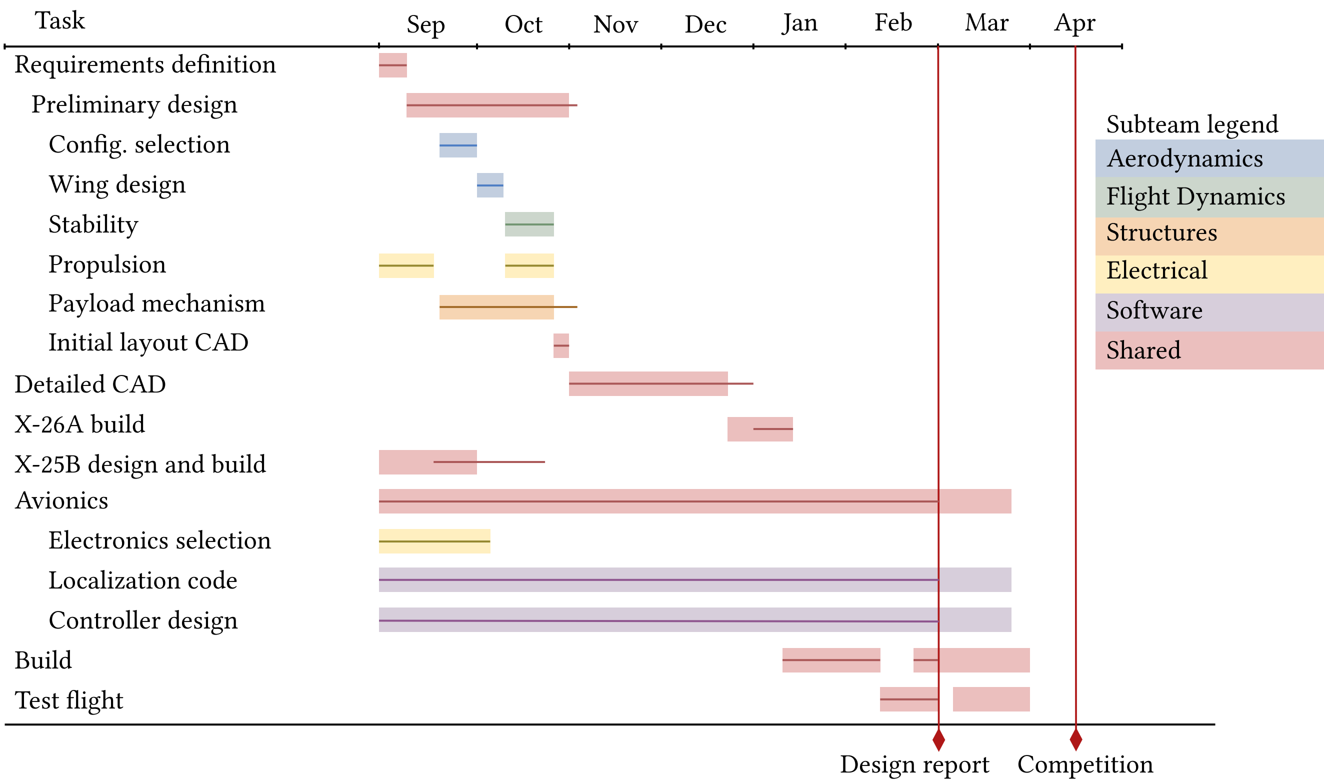


Figure 1.3 Design schedule divided by subteam. Shaded bars represent the planned schedule, solid lines are as-completed.

2 Aircraft Design

2.1 DESIGN TOOLS

The majority of the team's analysis tools are internal, home-grown codes that have been developed and matured over the past four years. Incremental progress has been made in improving the fidelity of these tools and cross-checking results using secondary methods. Open-source software and sponsored CAD/CAE packages are used for remaining analyses, as listed in Table 2.1

Table 2.1 Design and mission planning tools used by the team.

Section	Module	Tools used	Description
2.4	Constraint analysis	In-house MATLAB code	Wing sizing based on mission requirements, based on [1], improved using a drag model from [2], and propulsion model from [3].
2.5	Wing MDO	In-house MATLAB code	Wing planform design, spar sizing, optimizing aerostructural weight.
2.5.1	MDO aerodynamics	In-house MATLAB code	Planform design based on nonlinear lifting line method as described in [4] and airfoil data generated with XFOIL.
2.5.2	MDO structures	In-house MATLAB code	Spar sizing and weight estimation using classical beam bending, as formulated in [5].
2.6.1	Tail sizing scissor plot	In-house spreadsheet	Horizontal tail stability and control tradeoff, based on [6].
	Longitudinal static stability	In-house MATLAB code	Tail incidence and CG sweep, using nonlinear lifting line aerodynamic data from XFLR5.
2.6.2	Dynamic stability	XFLR5	Open-source software, using methods from [7].
2.6.3	Control surface sizing	In-house spreadsheets	Sizing for control authority requirements, based on [7], [8], and [6].
2.6.4	Servo sizing	XFOIL	Aerodynamic hinge moment calculation.
2.7	Propulsion selection	In-house MATLAB code	Motor-propeller analysis using methods from [3], with motor data from APC.
		RCbenchmark Software	Motor-propeller testing to determine thrust and current draw when used with a Tyto Robotics thrust testing stand.
2.8	Takeoff and climb-out performance	In-house MATLAB code	Simulation using velocity-dependent thrust/drag in an Euler time-march scheme.
2.9	V-n diagram	In-house MATLAB code	Safe operating envelope calculation, based on [1].
	Landing gear sizing	In-house MATLAB code	Landing load calculation, based off [9] and [10].
2.10	CAD	SOLIDWORKS 2024	Preliminary and detailed design.
3.1.2	Autopilot firmware	PX4 Autopilot	Low-level state estimation and autonomous flight control.
	Mission software framework	Robot Operating System	Coordination of software modules.
	Aircraft simulation software	Gazebo with advanced aerodynamics module	Complete mission simulation and validation of autonomy algorithms.
3.4	PCB design	Altium	Schematic drawing and PCB design for custom avionics.

2.2 SCORING ANALYSIS

Although completing more mission segments opens up more scoring opportunities, doing so requires additional avionics, shown in Table 2.2, that increase the aircraft's empty weight. To remain under the 3.5 lb weight limit, the payload weight must decrease as the empty weight increases, which reduces each segment score. Figure 2.1 shows flight scores as a function of the empty weight for four candidate mission profiles and autonomy levels, however all profiles were considered in the analysis. An autonomous delivery and capture mission provides the highest score when the empty weight is below 3.4 lb. Above this threshold, the payload penalty from the added systems outweighs the additional mission segment scores, and an autonomous release-only mission provides more points due to the additional payload afforded by removing the components needed to complete a payload capture. Conceptual design estimates for the chosen configuration described in Section 2.3 include the full avionics packages.

Table 2.2 Assumptions for scoring analysis.

Conops	Components required	Time bonus
Autonomous takeoff and landing	Empty airframe + 1 motor	2
Autonomous takeoff and landing, manual release	Above + camera, video link, payload mechanism (+0.12 lb)	2
Autonomous takeoff, landing, and release	Above + companion computer (+0.10 lb)	2
Autonomous takeoff, landing, delivery, and capture	Above + 2 motors (+0.28 lb)	1

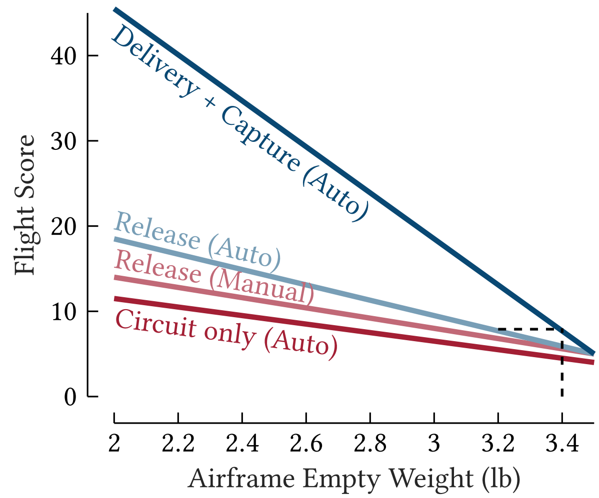
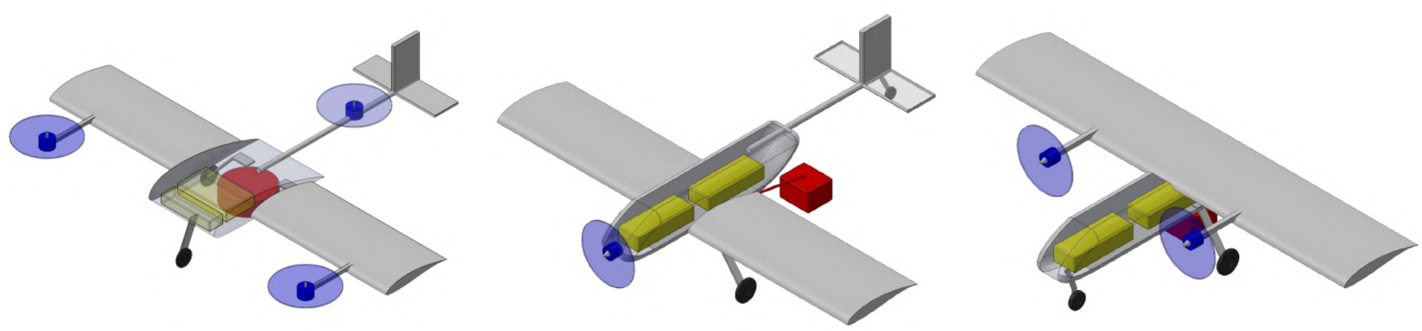


Figure 2.1 Flight score for four incremental levels of mission completion limited by avionics constraints.

2.3 CONFIGURATION SELECTION

A range of conceptual designs were synthesized. The mission score depends primarily on the empty weight, payload weight and the mission and development risks. Based on the substantial point bonus associated with fully autonomous mission execution, as identified in Section 2.2, it was assumed that all phases of the mission would be executed autonomously.

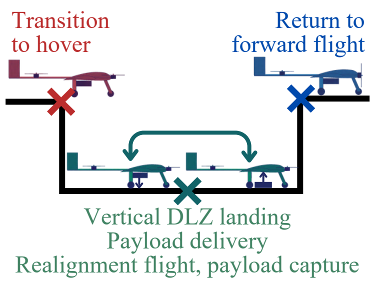
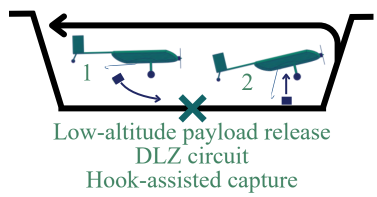
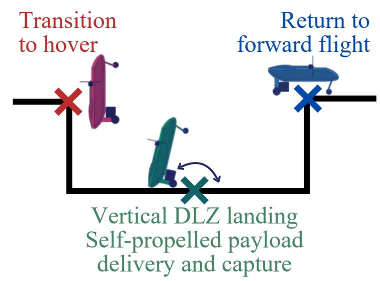
Three concepts were considered in the final concept evaluation, as presented in Figure 2.2 and evaluated in Table 2.3. The lifting body tricopter design was selected for its low mission risk, high technology readiness level (TRL), its similarity to previous years' designs and the availability of flyable prototypes.



(a) Lifting body tricopter (b) Conventional STOL (c) Flying plank tailsitter

Figure 2.2 Top three concepts evaluated. *Blue = Motors, Yellow = Avionics, Red = Payload*

Table 2.3   Airframe concept selection process.

Concepts	Lifting body tricopter	Conventional STOL	Flying plank tailsitter
Layout and payload description	Lifting body, three motors, two wing-mounted tiltrotors External payload capture and delivery mechanism	Tube and wing layout, single nose-mounted motor External hook payload capture and release *(+1 time bonus)*	Flying plank layout, two pivoting wing mounted motors External payload capture and delivery mechanism
Mission execution			
Empty weight	3.00 lb (Datum)	2.85 lb (-0.15 lb)	2.65 lb (-0.35 lb)
Mission risk factor (0-1)	1.0 - Datum	0.6 - Susceptible to wind and piloting accuracy	0.7 - Weak stability and susceptible to wind in hover state
TRL (1-9)	6 - Prototyped in vertical flight	8 - Proven in previous years	3 - Unproven controller/design
Risk-adjusted flight score	11.7	10.5	6.3

2.4 CONSTRAINT ANALYSIS

In the constraint analysis, the wing area was minimized to meet the MTOW = 3.5 lb requirement along with four flight phases including turn, climb, takeoff, and stall. The relationship between MTOW and wing area was derived using classical analyses [1], incorporating refinements such as accounting for viscous drag due to lift and utilizing a full motor and propeller thrust table. In Figure 2.3, the chosen wing area to meet the 3.5 lb weight limit was 1.83 ft^2. Stall is the limiting flight phase due to the large amount of thrust generated by the dual propeller configuration.

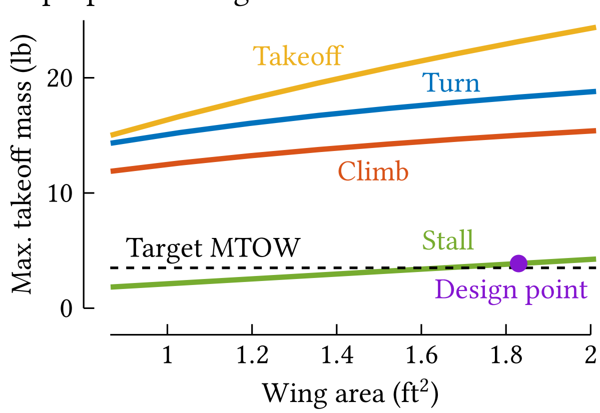


Figure 2.3   Enhanced constraint analysis model showing the relationship between MTOW and wing area for turn, climb, takeoff, and stall phases.

2.5 WING MULTIDISCIPLINARY DESIGN OPTIMIZATION

To design the wing outer mold line and size the spar, a MATLAB-based in-house multidisciplinary optimization framework was used to help conduct a trade study between a wing with and without a constant chord. A gradient-free surrogate optimizer[1] was selected to explore trade-offs between aerodynamic and structural performance. As the flight score heavily depends on payload capacity, the MDO maximizes MTOW - W_{struct}. The optimization framework can be seen in Figure 2.4.

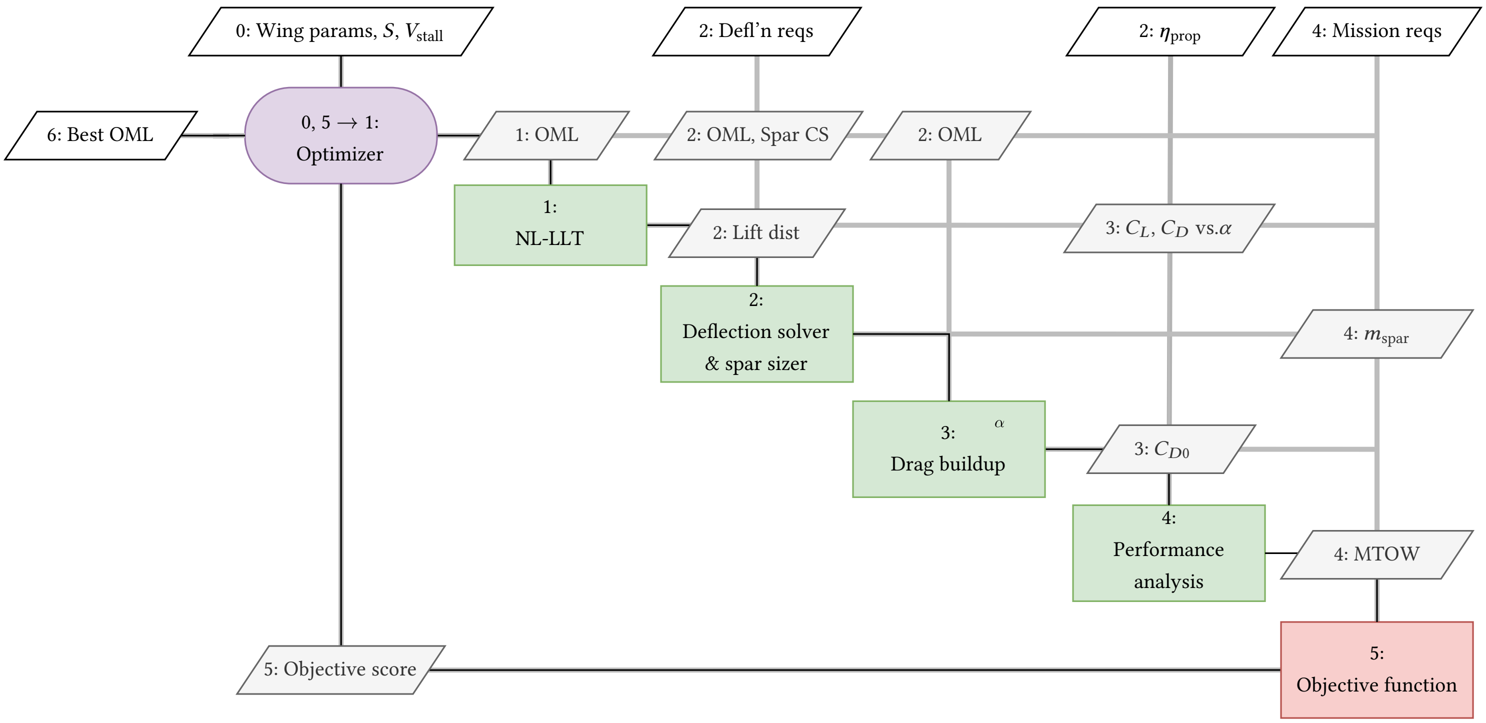


Figure 2.4 Design structure matrix of the wing MDO process, following the convention of [11].

2.5.1 *Aerodynamics*

To select the airfoil for the MDO analysis. The S1223, MH114, and Clark Y airfoils were compared. The S1223 has the highest $C_{L,\text{max}} = 2.23$ which can minimize wing size. However, its high $C_{L,\text{max}}$ makes it hard to stabilize in spiral mode. In contrast, the MH114 has a lower $C_{L,\text{max}} = 1.75$ but its thicker trailing edge simplifies manufacturing. Similarly, the Clark Y has a thicker trailing edge, though its low $C_{L,\text{max}} = 1.39$ causes the constraint analysis to increase the wing area to 2.37 ft^2 to meet the MTOW requirement, ultimately increasing weight. Therefore, with a thick trailing edge easing construction, and a moderate $C_{L,\text{max}}$ which minimizes weight and simplifies the stability analysis, the MH114 airfoil was selected.

2.5.2 *Structures*

The MDO minimizes the weight of a square pultruded CFRP tube to meet loading and deflection requirements (Section 1.1) based on the aerodynamic loads computed by the MDO. The bending and torsional

[1] www.mathworks.com/help/gads/surrogate-optimization-algorithm.html

loads and deflections of the CFRP tube located at the quarter-chord are calculated using classical beam bending theory [5]. A load factor of 2.85 derived from Section 2.9 was used along with an isotropic material model.

2.5.3 *Results*

A trade study was conducted by evaluating the MDO with and without a constant chord constraint. The constraint was introduced to simplify manufacturing. Although both cases satisfy the MTOW and structural requirements (ST-04a and ST-04b), the tapered wing more closely matches the elliptical lift distribution (Figure 2.5). However, the simplicity of the rectangular wing promotes greater manufacturing consistency, minimizing the number of structural and design inaccuracies. As the C_L–C_D performance is similar between the two cases (Figure 2.7), the rectangular wing was selected for the final design.

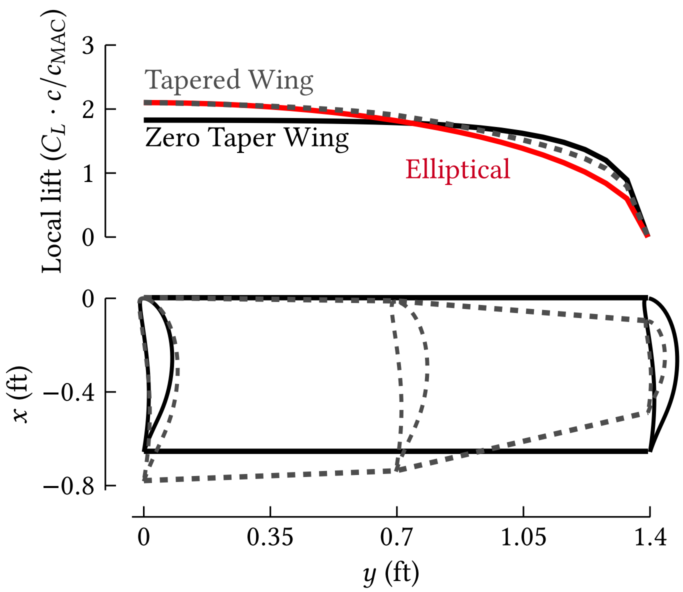


Figure 2.5 MDO aerodynamic results during cruise.

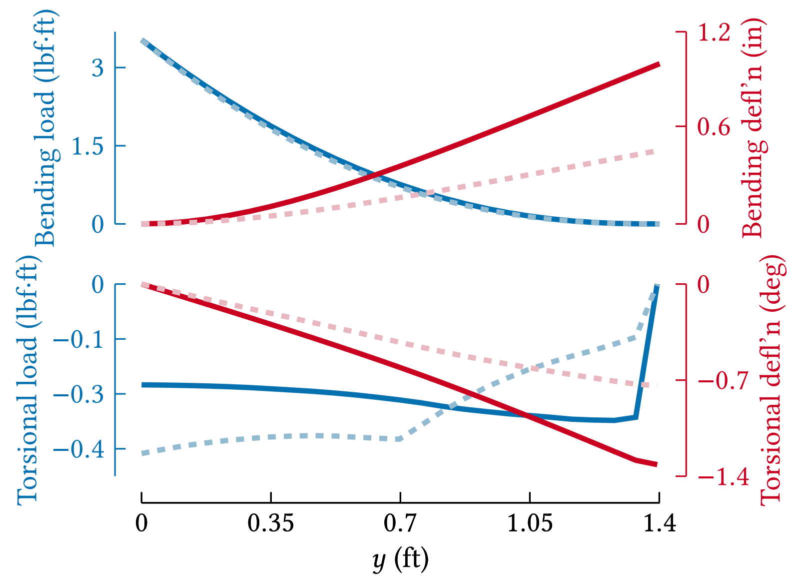


Figure 2.6 MDO structural results for tapered (dashed) and untapered (solid) wings under steady load.

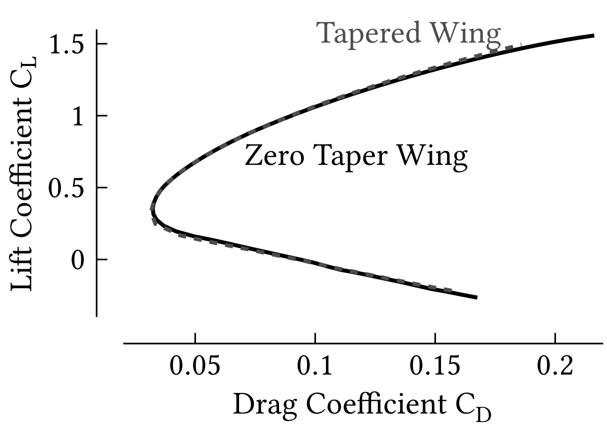


Figure 2.7 MDO C_L-C_D curve comparison.

Table 2.4 Wing MDO design results.

Parameter	Requirement	Design
Reference MAC		0.66 ft
Aspect ratio		4.25
Wing area		1.83 ft^2
MTOW	3.50 lb	3.83 lb
Stall speed	32 ft/s	32 ft/s
Max. climb gradient	10°	40°
Max. bank angle	45°	82°

2.6 STABILITY AND CONTROL

2.6.1 *Tail sizing and static stability*

The vertical tail was sized per [7]. Directional static stability ($C_{n\beta} > 0$), including a 10% safety margin for wake interference induced by the immediately upstream rear motor, is achieved with a vertical tail volume coefficient of 0.075 and $S_v/S_w = 0.11$. The scissor plot in Figure 2.8 estimates the horizontal tail size to achieve 5-15% static margin with a volume coefficient of 0.82 and Sh/Sw of 0.27. The aircraft can be trimmed across all operating conditions with a 5° deflection margin, satisfying requirement SC-08.

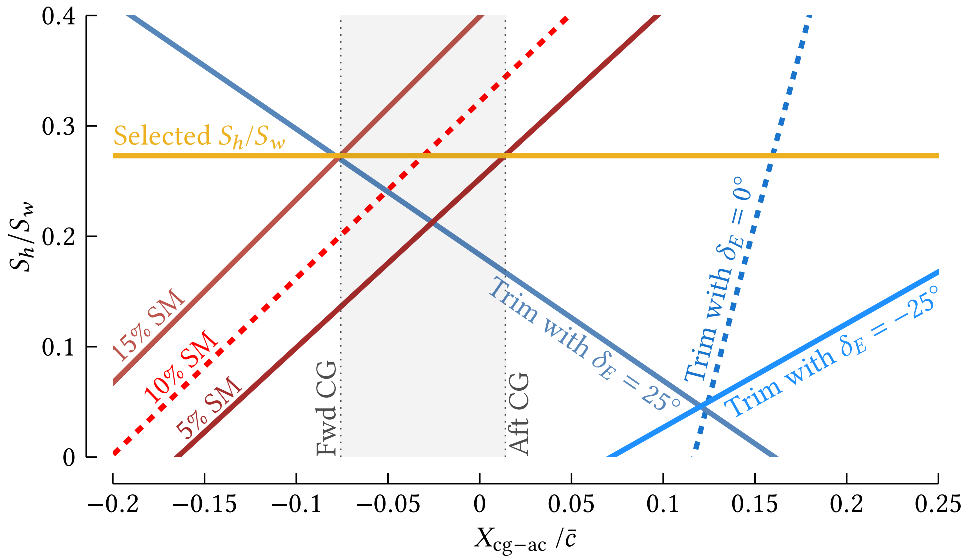


Figure 2.8 Linear tail sizing under worst-case trim conditions (V_{stall}).

Subsequent analysis accounting for viscous aerodynamic effects verified that the desired static margin is achieved when the center of gravity is positioned between 0.17c and 0.25c aft of the wing leading edge (Figure 2.9). As shown in Figure 2.10, a tail incidence angle of −2° trims the aircraft in cruise flight without elevator inputs and remains trimmable in all flight conditions.

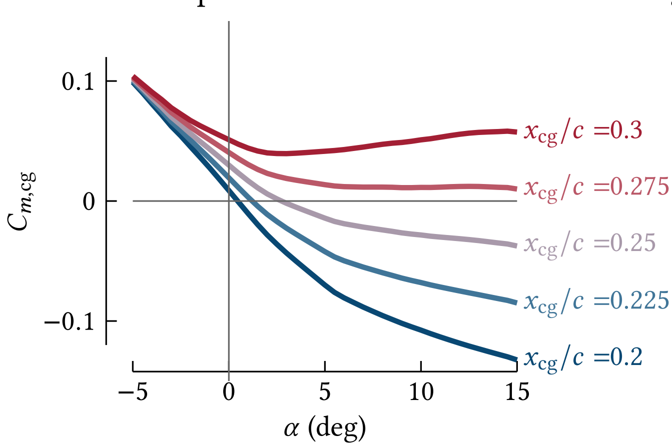


Figure 2.9 Center of gravity placement using viscous aerodynamic performance.

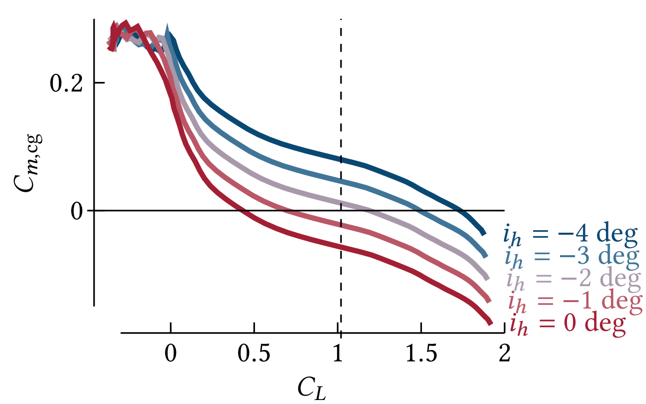


Figure 2.10 Selection of the tail incidence angle, i_h.

2.6.2 *Dynamic stability*

Dynamic stability analyses were conducted for three loading conditions ($m_{\text{OEW}} = 2.84$ lb, $m_{\text{mid}} = 3.17\ lb$, and MTOW). Sweeps of trimmed elevator deflections were conducted across three CG positions spanning the range determined in Section 2.6.1 using the vortex lattice method in XFLR5. As shown in Figure 2.11 and Table 2.5, all eigenvalues meet the dynamic stability criteria in [12], except for the spiral mode at low speeds. For $V > V_{\text{climb}} = 36.7$ ft/s, the spiral mode doubling time satisfies stability requirement (SC-05). Reduced spiral stability at low speeds was deemed acceptable, provided extended low-speed unstabilized operation is avoided.

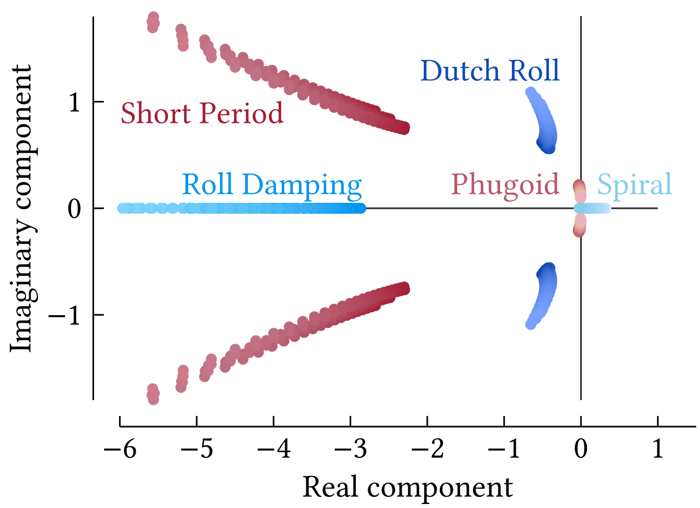


Figure 2.11 Root locus at three CG locations.

Table 2.5 Dynamic stability performance.

Mode	Requirement	Worst Case
Spiral	Doubling time > 20 s	2.14 s
Dutch roll	Damping ratio > 0.08	0.082
Roll damping	Time constant < 1.0 s	0.35 s
Phugoid	Damping ratio > 0	0.005
Short period	Damping ratio 0.35-2.0	0.42

2.6.3 *Control surface sizing*

The ailerons, elevator, and rudder were designed according to [8], [6], and [7], respectively. The associated design requirements and resulting control surface sizes are shown in Table 2.6.

Table 2.6 Control surface design inputs and outputs.

Control surface	Requirement	Size and configuration
Aileron	Bank to $\phi = 60°$ in <1 s at 1.13 V_{stall} (SC-06)	30% chord, 20° deflection, spanning from 0.6b to 0.9b
Elevator	Achieve a runway pitch rate of 30°/s at V_{stall} (SC-07) Trim at V_{stall} with a 5° deflection margin (SC-08)	35% chord, 30° deflection, spanning full horizontal stabilizer
Rudder	Maintain directional control in single front-motor operations (SC-09)	40% chord, 15° deflection, spanning full vertical stabilizer

2.6.4 *Servo sizing*

The maximum hinge moments for each control surface were determined by simulating their respective airfoils and dimensions with hinged control sections in XFOIL. The worst-case C_p distributions over the deflected control surface area were integrated to compute the total hinge moment. The analysis assumed uniform pressure loading across the span of the control surface. The maximum torque required for each control surface was well below the rated servo torque of 4.0 lb·in, as shown in Table 2.7.

Table 2.7 Servo sizing inputs and results.

Control surface	Torque (lb·in)	Conditions
Aileron	0.496	20° wing α, 20° deflection, V_{max}
Elevator	0.673	18° h-stab α, 30° deflection, V_{max}
Rudder	0.195	20° β, 15° deflection, V_{max}

2.7 PROPULSION

Propulsion selection was conducted under the hard constraint of propeller sizes under 5.1 in brought about by requirement PR-04. The primary requirement for the propulsion system is a 1.3 T/W (PR-02) [8] given an aircraft weight of 3.5 lb (MC-01). Four motors that satisfy the aforementioned requirements while minimizing weight and power consumption were selected based on manufacturer spec sheet data. An in-house script was then used to simulate three propellers against four motors to obtain propulsion efficiency values given our mission parameters. For brevity, only the top performing motor, the T-HOBBY F2203.5 2850 K_v's results are displayed in Figure 2.12 and Figure 2.13. Table 2.8 shows experimental static thrust testing results for this motor. Discrepancies in measured efficiencies can be attributed to drops in battery voltage and vibrations. Given these results, the T-HOBBY F2203.5 2850 K_v paired with the 5.0" × 3.7" propeller is chosen.

Table 2.8 Experimental values across different propellers for selected motor based on a 1.3 T/W

Prop size (diameter × pitch)	5.0" × 3.7"	5.0" × 4.0"	5.1" × 5.0"
Current (A)	13.8	20.9	12.1
$\eta \times 10^3$ (lb/W)	5.99	5.2	3.51

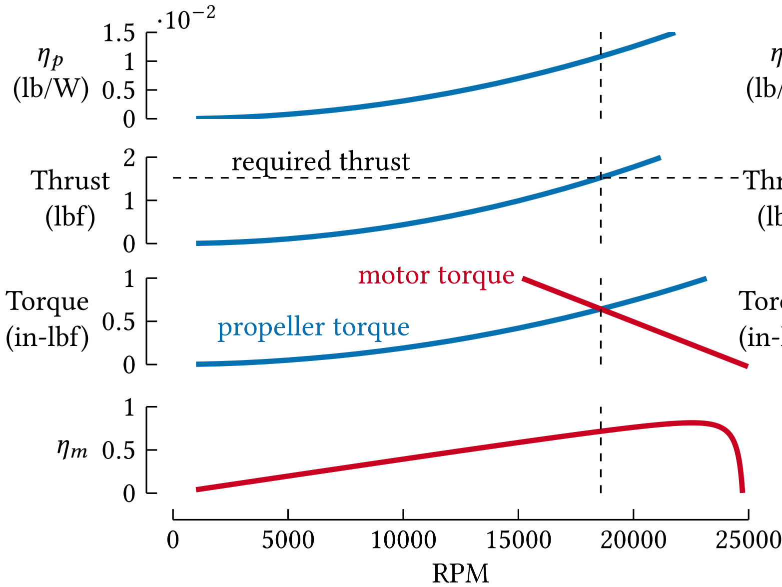


Figure 2.12 Motor-propeller matching for static thrust (hover with 1.3 T/W)

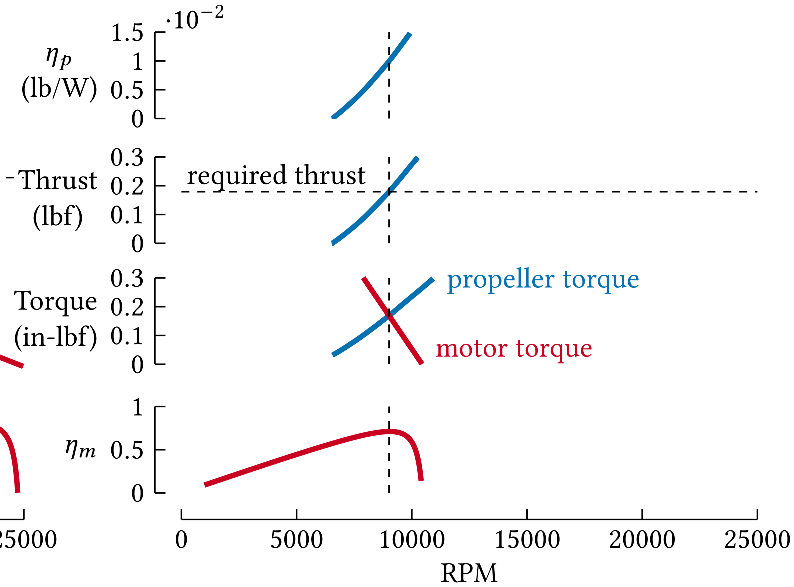


Figure 2.13 Motor-propeller matching for dynamic thrust (steady level flight)

2.8 Takeoff and climb-out performance

Thrust and drag variation with airspeed were modeled using a forward time-integration approach to simulate takeoff and climb performance. The equations of motion were integrated using an explicit Euler scheme with verified convergence at a 0.02 s time step. As shown in Figure 2.14, takeoff is achieved within 25 ft. The maximum climb performance at $1.13V_{\text{stall}}$ corresponds to a climb angle of 32°.

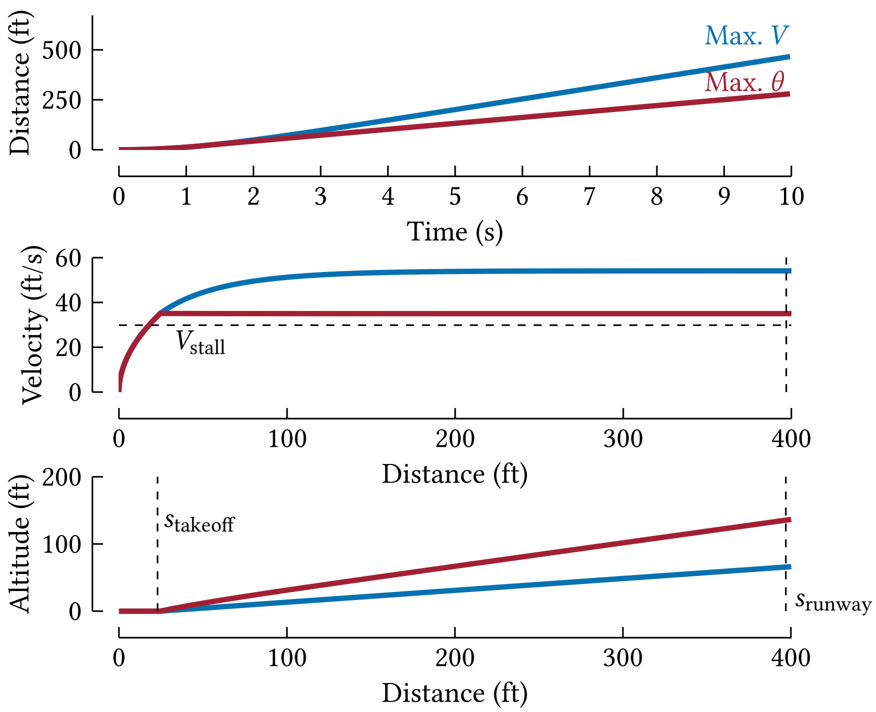


Figure 2.14 Takeoff and climb-out simulation.

2.9 DERIVED FLIGHT LOADS AND STRUCTURAL TESTING

The static wing loads are determined from the prediction of the load factor in the expected flight conditions shown in Figure 2.15. A maximum load factor of +2.85 defines the flight envelope at the max maneuvering speed of 54.8 ft/s, and gust speed of 20 ft/s [13].

Table 2.9 summarizes the critical loads of the primary structures, and shows how all requirements are satisfied. Representative static and dynamic load testing examples are shown in Figure 2.16. All structures were governed by their associated maximum deflection requirements, rather than material failure limits.

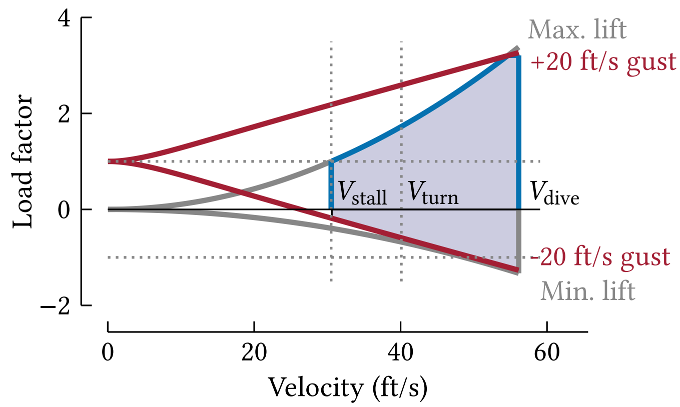
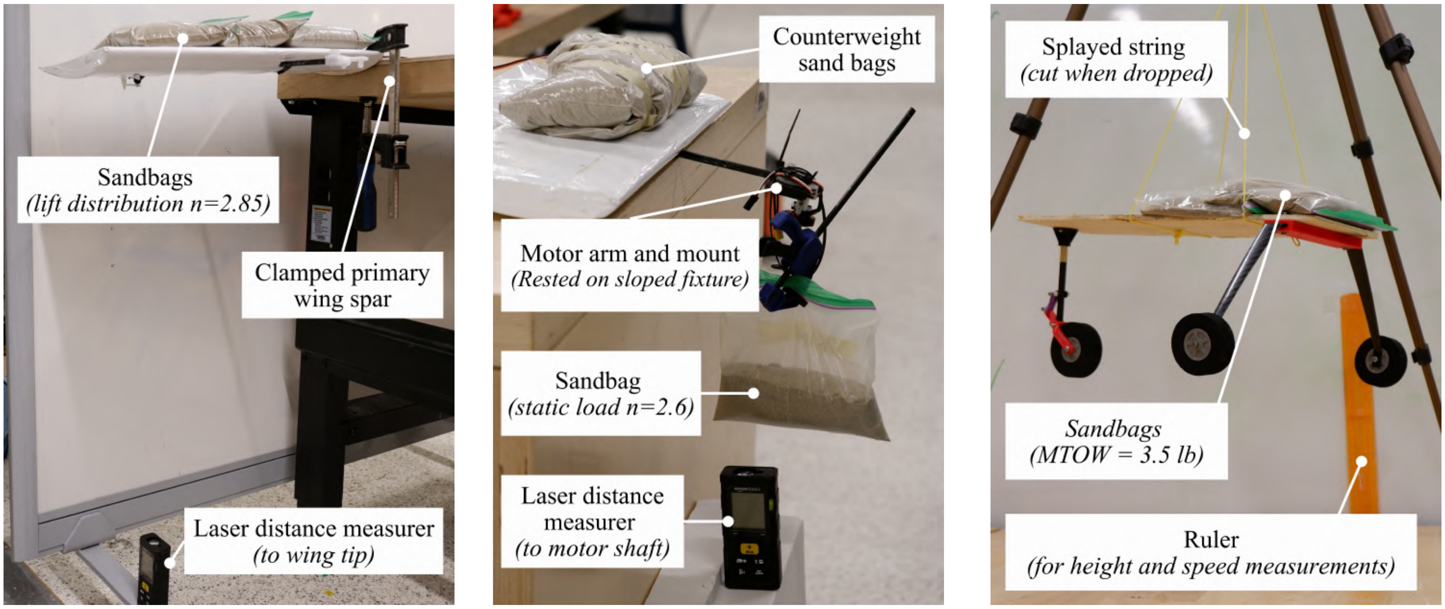


Figure 2.15 $V-n$ diagram.

Table 2.9 Summary of loads and satisfaction of requirements.

Component	Case	Loading Conditions	Design Load	Analysis Methods	Test result
Wing	V-n limit	Shown in Figure 2.15 V_{max} = 55 ft/s V_{gust} = 20 ft/s	Lift and moment distribution n=2.85	Classical beam analysis FEA Static load test	Tip bending deflection: 1.5 in Tip torsion deflection: 1.9° Failure safety factor: 2.0
Motor arm	Vertical ascent	T/W = 1.3 Margin of safety = 2.0	Static n=2.6	Classical beam analysis Static load test	Motor deflection: 0.12 in Failure safety factor: 2.1
Main landing gear	Static load	Taxi bump with aft CG [9]	Static n=2.0	FEA Static load test	Vertical axle travel: 0.47 in
	Dynamic load	Hard taildown landing at MTOW $V_{vertical}$ = 6.5 ft/s [9]	Dynamic n=3.13	Hand calculations Vertical drop test	Vertical axle travel: 2.5 in Failure safety factor: 3.2
Nose landing gear	Static load	Taxi bump with forward CG [9]	Static n=2.25	Hand calculations Static load test	Vertical axle travel: 0.35 in
	Dynamic load	Hard three-point landing at MTOW $V_{vertical}$ = 6.5 ft/s [9]	Dynamic n=3.13	Hand calculations Vertical drop test	Vertical axle travel: 0.51 in Failure safety factor: 2.1



(a) Wing (b) Motor arm (c) Main landing gear

Figure 2.16 Static and dynamic load tests as carried out for various components.

2.10 LAYOUT AND MANUFACTURING

Key factors considered in the materials selection and manufacturing methods are shown in Table 2.11. The majority of the airframe is constructed of balsa wood, which is laser cut in the university's onsite fabrication facility and joined instant cure cyanoacrylate glue. Carbon fiber is used to reinforce and reduce the material needed in key structural components. 3D printed parts are used to facilitate the production of tolerance-critical complex geometries, such as the motor, tail, and wing mounts. The material selections are shown in Figure 2.17.

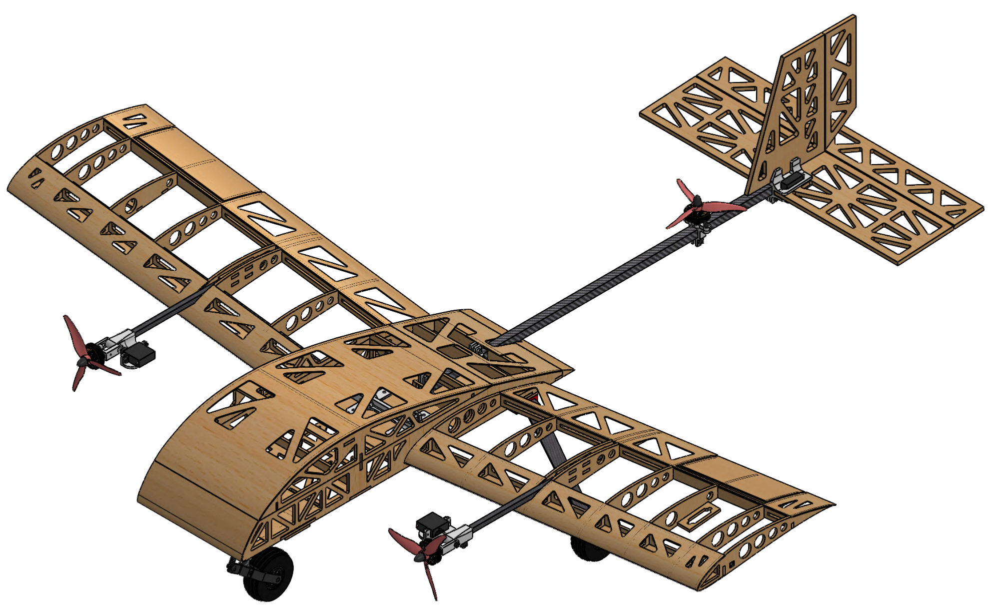
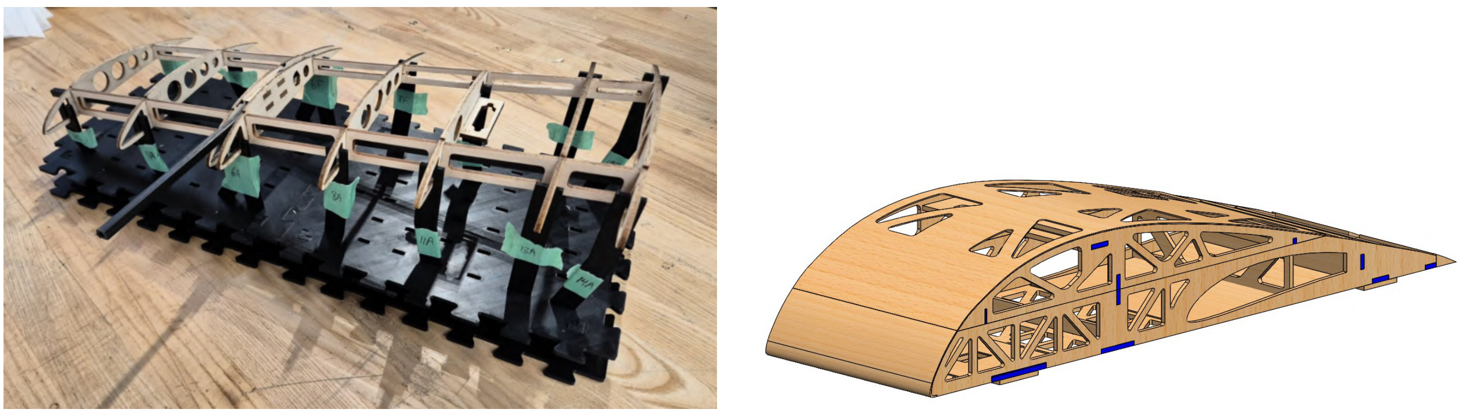


Figure 2.17 Detailed design of UT-26 with skinning hidden, showing material selection.

The wings are manufactured using a custom 3D printed jig to ensure vertical and spanwise alignment, as shown in Figure 2.18a. Chordwise alignment is provided by the primary CFRP spar. The horizontal and vertical stabilizers consist of single-piece flat balsa sheets, removing the need for seperate tail ribs and spars and improving manufacturability. The control surfaces are attached with composite hinge sheets, which simplifies alignment, and reduces the risk of binding. Self-jigging elements and alignment tabs are integrated throughout the fuselage to ensure a <1/32 in tolerance, as shown in Figure 2.18b. The airframe is skinned with a heat-shrink adhesive film to maintain the aircraft form and create a smooth surface finish. The airframe part count was minimized to 83 components and fasteners were standardized to streamline assembly and repair, enabling <10-minute assembly/disassembly with two screwdrivers, per requirement MA-02. Disassembled, the aircraft fits in a 22 x 14 x 14 in box.



(a) Wing jig

(b) Fuselage alignment tabs (shown in blue)

Figure 2.18 Manufacturing methods and tools used to assemble the UT-26 airframe.

Table 2.10 Aircraft mass balance.

Aircraft component	x-location from wing $c/4$ (in)	Mass (lb)	Aircraft component	x-location from wing $c/4$ (in)	Mass (lb)	Summary	
Airframe			*Propulsion*			DAS weight	0.588 lb
Wing	0.00	0.331	Forward motors + props	-4.33	0.095	Airframe weight	1.259 lb
Fuselage	-4.15	0.331	Rear motor + prop	18.29	0.049	Propulsion weight	1.228 lb
Tail	23.62	0.130	Battery	-6.67	0.597	Empty weight	3.045 lb
Tail boom	13.50	0.108	BEC	-1.40	0.108	Total weight	3.500 lb
MLG + wheels	6.35	0.127	ESC	-3.05	0.093		
NLG + wheel	-6.42	0.046	Radio receiver	1.78	0.011	Estimated cg behind $c/4$	-0.095 in
						NP location behind $c/4$	0.630 in
DAS			*Payload*				
Camera	-0.43	0.013	Payload mechanism	0.00	0.110		
Lidar	-2.83	0.051	Payload servo	0.00	0.026		
GPS	-1.03	0.095	Payload	0.00	0.220		
Pixhawk + carrier	-5.05	0.167					

Table 2.11 Material selection considerations.

Material	Benefits	Drawbacks	Applications	Key Design Driver
Balsa	Lightweight (~8.74 lb/ft^3) Laser cutting produces accurate (±1/1000 in) parts	Poor load-bearing capabilities (~3.00 ksi modulus of rupture) Low bending stiffness (~500 ksi flexural modulus)	Wing ribs, sheeting, secondary bulkheads	Secondary structures with low loading and required stiffness Minor structures safety factor to yield (ST-01b)
Plywood	Laser cutting produces accurate (±1/1000 in) parts More durable than balsa	Heavier than balsa (~28.10 lb/ft^3) Laser cutter maximum thickness <1/8 in	Primary bulkheads, wing root and tip ribs, secondary wingspars	Major structures with higher loading or stiffness. Major structures safety factor (ST-01a)
Carbon fiber	Very high stiffness-to-weight ratio (~370Msi specific modulus)	Very high cost Labour-intensive manufacturing Attenuates R/F signals Difficult to repair	COTS landing gear Wingspars, tailboom, motor arms	Max. wingtip deflection (ST-04a, ST-04b) Max. tailboom deflection (ST-05)
3D printed plastics	Medium-to-high flexibility (~300ksi flexural modulus) FDM 3D printing produces accurate (±2/1000 in) parts Complex geometries supported	Low strength-to-weight ratio (~200ksi/(lb/in^3) specific strength) Long print times for large parts	PETG: Tailboom-tail, wing-fuselage interfaces, motor mounts PLA: Payload	Maximum assembly time (MA-02) Maximum gap size at subassembly interfaces (MA-03)
Expanded polymer foams	More impact resistant than wood Very low cost	Wire cutting is not well suited to complex wing geometries Poor surface finish Heavier than balsa structures Difficult to repair	Prototyped during preliminary design Not used in final design	No manufacturing benefits Increased weight compared to skinned balsa frame

3 Mission Objectives

3.1 Overall Mission Plan and Tradeoffs

3.1.1 *Mission sequence*

The mission sequence is triggered by the Payload Operator and then proceeds autonomously, as shown in the TDS. The vehicle takes off, performs a precision landing on the DLZ using GPS and computer vision, then returns to base as shown in Figure 3.1.

The overall mission plan was designed based on four main requirements: 1. Landing accuracy: The vehicle must land within 2 in to perform payload capture; 2. Speed: The mission sequence shall be as short as possible; 3. Weight: Components shall be lightweight to maximize payload capacity; 4. Reliability: The mission sequence shall be repeatable, simple to implement, and minimize potential failure points.

This multi-stage landing approach eliminates the need for a search pattern and reduces mission time by ~30 s while improving landing accuracy. Since electronics are now permitted on the payload, trade studies are summarized in Table 3.1 to further support the use of computer vision as a method of guidance.

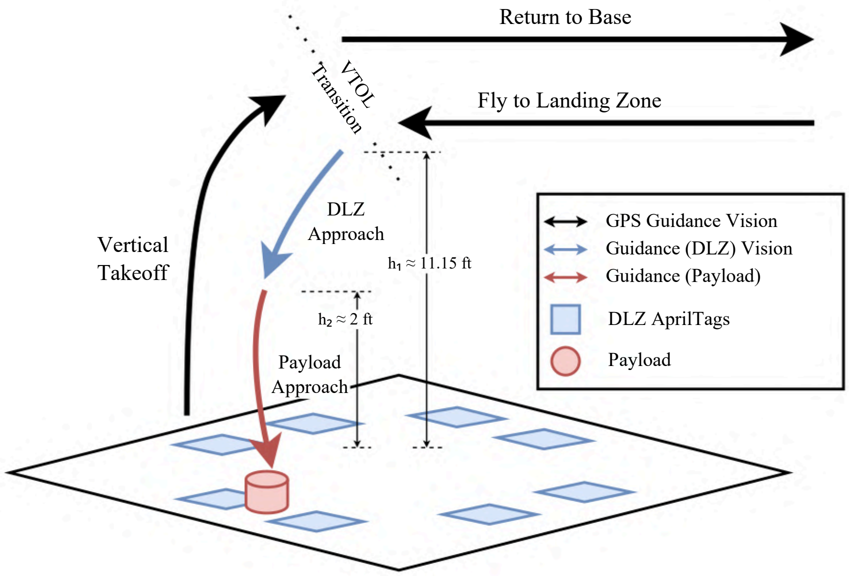


Figure 3.1 Summary of the overall mission sequence.

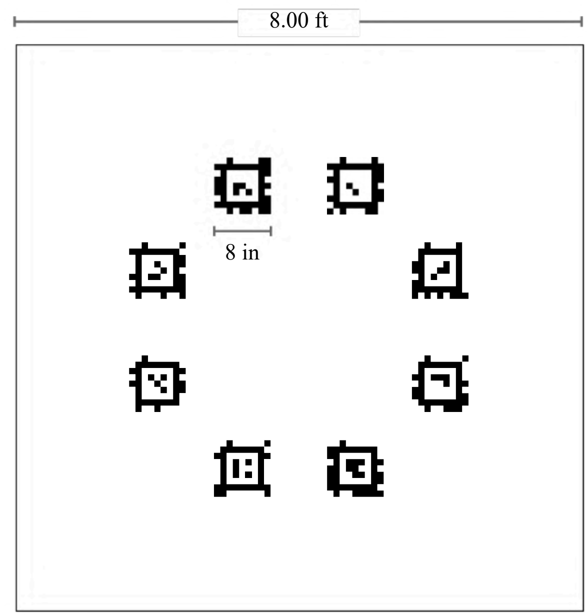


Figure 3.2 Layout of AprilTags on the DLZ.

Table 3.1 Summary of trade studies for vehicle guidance system.

Guidance method (chosen in *italics*)	Rationale for choice
GPS only	GPS alone is not accurate enough (~8 in) to land on the payload.
Computer vision	CV has good accuracy (~2 in) and it only requires a lightweight camera (~0.013 lb) and an onboard computer. Imposes little requirements on payload geometry and mass. It also allows for various detection methods and filtering techniques that are well supported and documented.
Autonomous driving payload	More components are needed, such as a power supply, wheels, additional processing module, sensors, motors and motor drivers. It substantially increases points of failure, and increases the minimum payload mass.
LiDAR	Requires a search pattern which can add an estimated 30 s or more to the mission sequence. May also be sensitive to the terrain and background noise which will change between the test flight field and the competition field.
Other sensors (IR beacon, RFID, UWB)	IR beacon may not perform well in outdoor conditions directly under sunlight. RFID has poor accuracy and is mostly used for identification purposes. UWB requires additional anchors on the DLZ to localize the payload, which is not permitted by competition rules.

3.1.2 *Software architecture*

Low-level state estimation and control are managed by a FCU running PX4 Autopilot[2], an open-source autopilot software selected for its robust flight performance and compatibility with our existing system. High-level mission is planned and executed by an onboard mission computer running Robot Operating System 1[3] (ROS 1), which facilitates communication between various components such as FCU, GCS, payload mechanism, and camera. ROS 1 was selected for its broad ecosystem of supported packages required for the mission, especially for computer vision based guidance. For early validation of the guidance algorithms, Gazebo[4] physics simulator was used with a software-in-the-loop (SITL) simulation

[2] px4.io/
[3] www.ros.org/
[4] gazebosim.org/home

of PX4 to model the complete mission.

3.2 PAYLOAD AND MECHANISM

The design was guided by the following requirements: 1. It must allow payload retrieval and capture, together or separately, while landed on the DLZ, 2. Lightweight: must be under 0.12 lb without actuators, 3. Accuracy: must account up to 2 in of landing inaccuracies, 4. Complexity: must not use more than 2 actuators, 5. Payload weight: must hold at least 0.22 lb of payload.

The mechanism, shown in Figure 3.3, uses a string with a magnet attached to secure the payload. The string is deployed and retracted from a spool, actuated by a 360° servo with an encoder. The servo is commanded by the onboard computer through the mission state machine and PID control loop. The payload is a cylinder with an inverted cone-shaped hole from the top surface. The hole self-centers the mechanism, allowing for landing inaccuracies up to 2 in. The payload is in a bright colour to emphasise the contrast with the background, improving the accuracy of the colour-based CV guidance system.

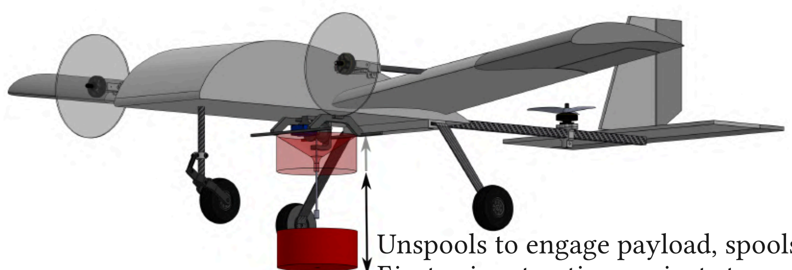


Figure 3.3 Payload mechanism operation.

3.3 COMPUTER VISION AND PRECISION LANDING

3.3.1 *Computer vision and localization*

The design of the CV localization system was guided by the following criteria: 1. Minimum detection ranges of 11.5 ft for DLZ and 3.3 ft for payload (based on camera FOV and expected GPS error); 2. Minimum detection update rate of 20 Hz; 3. Minimum localization accuracy of 2 in to facilitate precision landing; and 4. Payload localization must not require a payload geometry with a flat top surface (to allow for self-centering and simple payload capture designs).

AprilTag [14] was selected as the ficidual marker for DLZ localization due to its robustness and range. Based on the measured GPS accuracy Figure 3.4 and now consistently achieving 3D RTK GPS lock, the required tag size was calculated as shown in Figure 3.2. The DLZ layout contains four Payload Zones, which are pairs of AprilTags the vehicle can choose to either drop or capture a payload. The positions and sizes of the tags shown in the figure accounts for the tested detection ranges and camera FOV.

Payload localization is achieved using a colour detection algorithm since it can account for non-flat surfaces and the small size of the payload. DLZ is black to reduce light reflected onto the payload, decreasing detection noise.

Flight attempt	Offset from target (in)
Flight 1	10.6
Flight 2	2.6
Flight 3	4.3
Flight 4	13.6
Flight 5	4.5
Flight 6	8.5
Flight 7	13.4
Average	8.2
Outlier (doubled flight time)	20.1

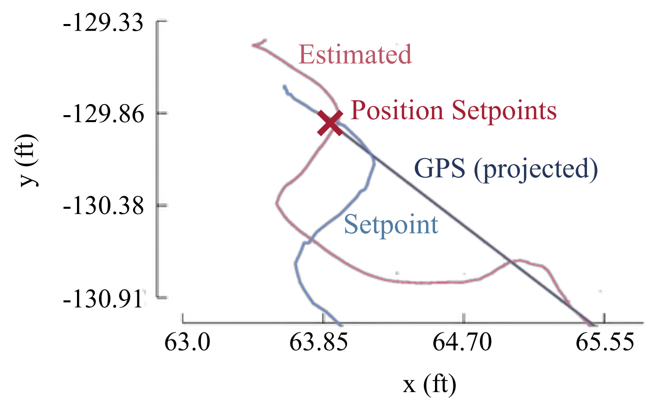


Figure 3.4 GPS test results with flight-by-flight offset from target (left) and an example flight log from flight attempt 6 (right). The red X indicates the commanded landing GPS coordinates

The `apriltag_ros` ROS package is used for Apriltag detections [15]. False positives are filtered by discarding measurements with large discontinuities in the estimated target pose. As in [16], a custom extended Kalman filter (EKF) is implemented to estimate the final target pose in a body-fixed frame, avoiding sensitivity to potential GPS and magnetometer yaw errors. AprilTag detections and a laser altimeter provide measurement updates, while state prediction relies on position estimates from PX4.

(a) DLZ AprilTag viewed from max range.

(b) Coloured payload viewed from max range.

Figure 3.5 Images from test footage showing the max range of the DLZ and payload AprilTags. The DLZ pictured represents a fraction of the competition spec.

The system was validated through a series of tricopter tests on X-26 Figure 3.5, which verified that all requirements were met. The DLZ AprilTag and colour detection ranges are 16.6 ± 2.3 ft and 6.2 ± 1.0 ft respectively, while maintaining 21 Hz detections at 70% CPU usage, and a localization accuracy within

2 in at close range.

3.3.2 *Precision landing*

The precision landing algorithm performs both payload release and capture through sending velocity and position setpoints to low level PX4 controllers. When the target is visible, the vehicle aligns itself horizontally to the target while descending at a rate proportional to its altitude. If the horizontal position error exceeds 1.5 times the altitude, vertical descent is paused to allow lateral correction, forming a volume of acceptance above the target as described in [17]. If the target is lost, the vehicle ascends to a predefined altitude and executes a search pattern until the target is visible. This high level precision landing controller was tested through Gazebo physics simulation, and low level attitude and rate controllers were tuned through test flights.

3.4 ELECTRONICS

UT-26 MAPLE's electronic schematic is shown in Figure 3.6. Table 3.2 lists the major design decisions and which requirement they target.

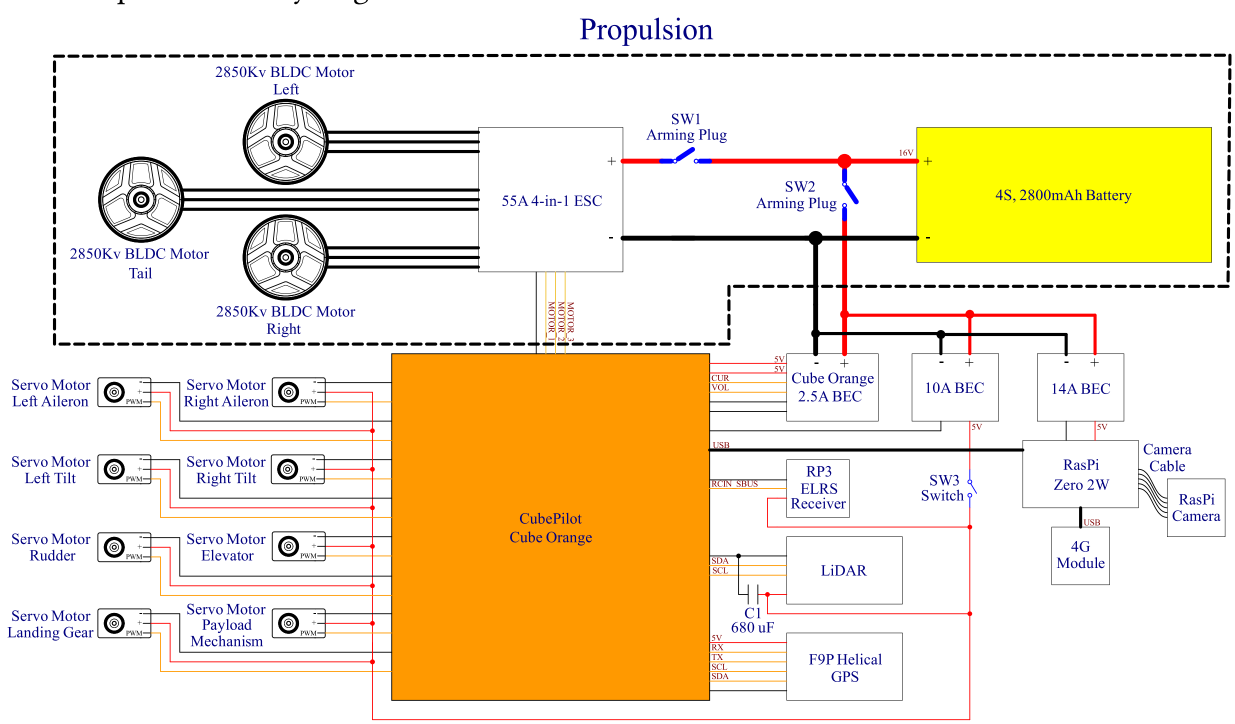


Figure 3.6 Aircraft circuit schematic.

Table 3.2 Major design decisions in UT-26 MAPLE's electronic system.

Requirement	Design Decision and Justification
MC-01 (Max. takeoff weight)	Competition rules allow for a separate battery back or a BEC to power the receiver system. The latter was used, saving 0.26 lbs of battery weight.
MC-01 (Max. takeoff weight)	A Raspberry Pi Zero 2 W was selected due to its small form factor (2.56 in × 1.18 in) providing a lightweight solution (0v024 lbs) for computer vision.
PR-03 (Max. battery capacity)	Empirical measurements from Section 2.7 was used to determine a 2612 mAh battery capacity of the propulsion system alone for 4 minutes of flight at 1.3 T/W. 202 additional mAh was calculated from fully loading all other avionics for 4 minutes. A 2800 mAh battery was chosen as a result.
All DAS requirements	To satisfy all DAS requirements, a separate BEC was chosen to prevent brown-out on the Raspberry Pi Zero 2 W by supplying a stable 5V, isolated from high-current draw avionics such as servo motors. A 4G LTE module is selected for out telemetry system due to its superior bandwidth and range (Table 3.3), and integrability with the Raspberry Pi Zero 2 W.

Table 3.3 Comparison of telemetry datalink alternatives.

Datalink	Max. range (mi)	Max. bandwidth (Mibps)	Latency (ms)	Weight with antennas (lbs)
SiK radio	0.06–0.19	0.115	50–150	0.062
RFD900x	3.11–24.85	0.115	50–100	0.073
Cellular 4G LTE	12.43 (from nearest cell tower)	50	100–200	0.031

4 Conclusion

UT-26 MAPLE has been designed and built by UTAT to serve as a suitable V/STOL tricopter for autonomous package deliveries and retrievals. Leveraging an airframe designed for assembly and manufacturing and a fully autonomous system to maximize mission performance, UT-26 MAPLE is expected to achieve a total flight score of 35.82. The use of matured and upgraded in-house tools enabled the analysis of the airframe, while the development of three prototype aircraft allowed for the rapid maturing of the software stack. Future work prior to the competition will focus on executing a comprehensive flight testing campaign and streamlining the manufacturing process to enable the production of additional airframes for testing and competition readiness.

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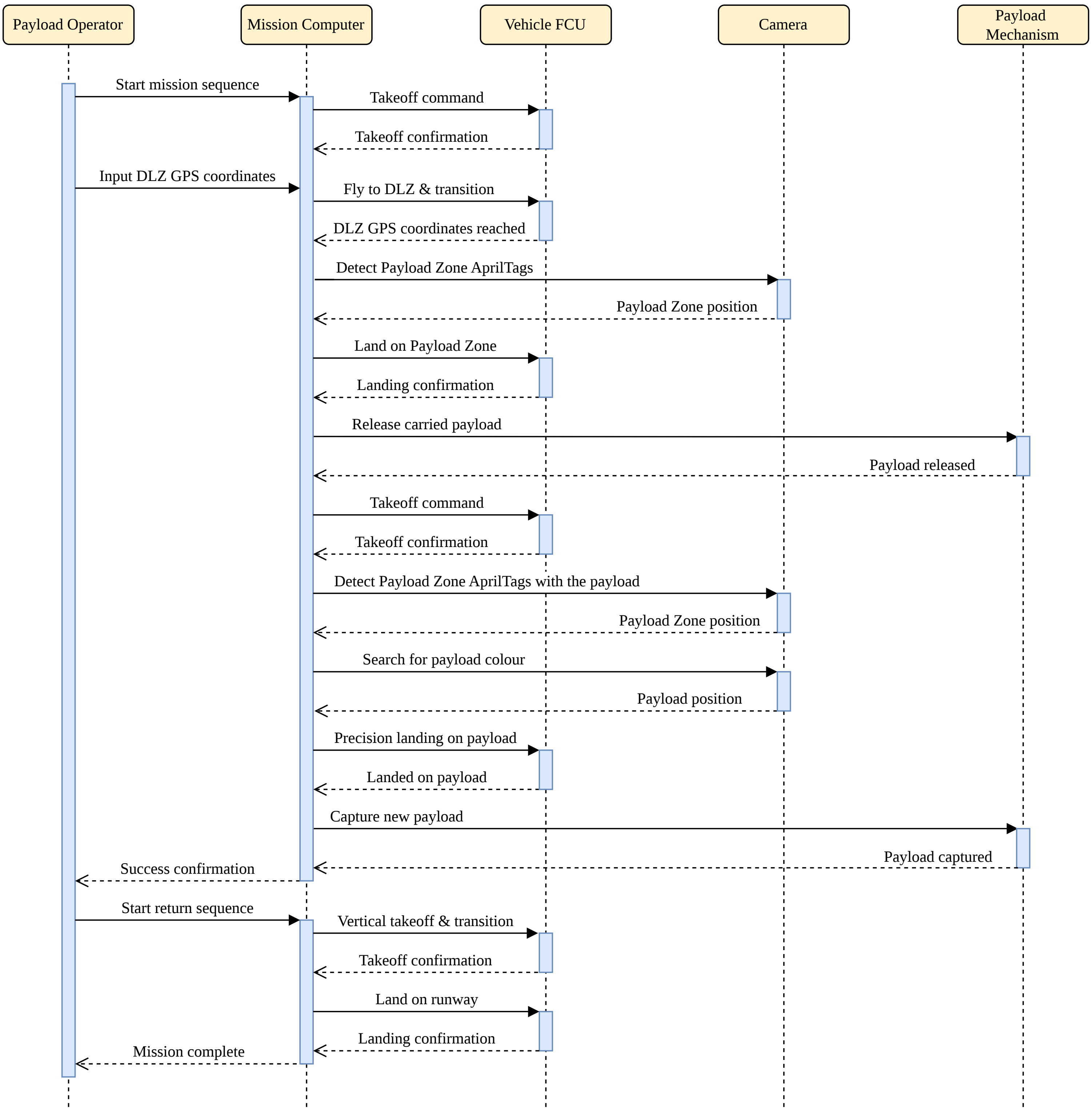
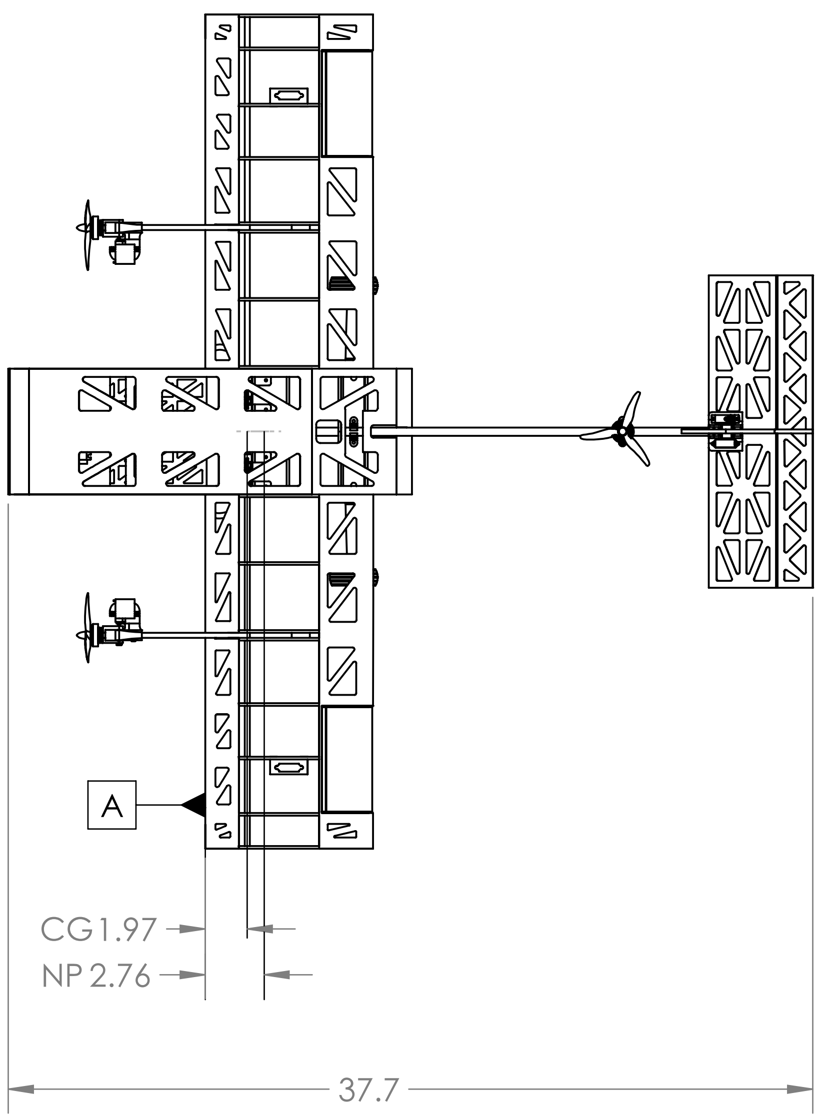
Sequence Diagram for Payload Delivery/Capture

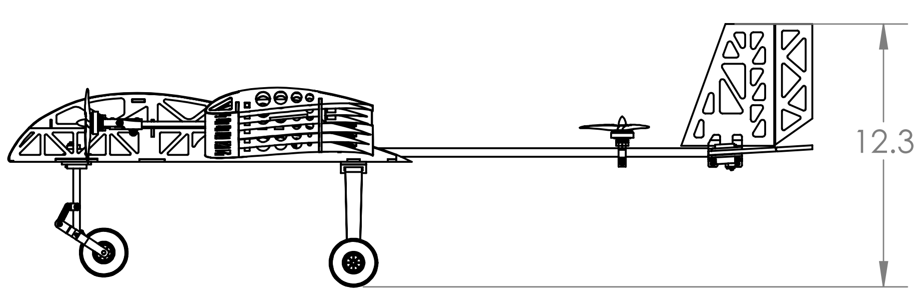
U of T Aerospace Team Team 206 Univ. of Toronto

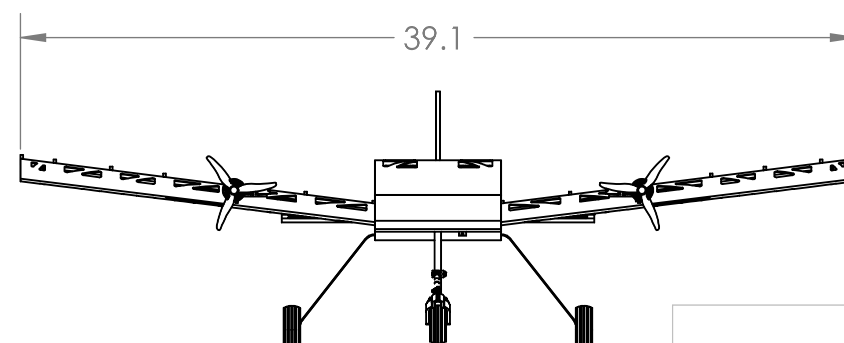
Roles: Operator: initiates mission sequence and provides DLZ coordinates; Mission Computer: plans & executes mission sequence; Vehicle FCU (flight control unit): low-level flight control; Camera: locates DLZ & payload for precision landing using AprilTags and colour; Payload Mechanism: releases carried payload or captures new payload. When initiated by the operator, the vehicle will perform a conventional takeoff, followed by a precision vertical landing on one of the empty Payload Zones guided by AprilTags. After releasing the carried payload, the vehicle will takeoff vertically and search for the Payload Zone with a different payload. Then the vehicle will precision land by the payload using AprilTags and colour. Once the payload is captured, the vehicle performs a vertical takeoff and returns to base with a conventional landing on the runway.



NOTE: PAYLOAD MECHANISM MAY BE REMOVED DEPENDING ON THE FLIGHT







WEIGHT AND BALANCE

ITEM	LOCATION FROM DATUM A (IN)	WEIGHT (LB)	MOMENT (LB*IN)
Motor-front (x2)	-4.3	0.087	-0.38
Motor-rear	18	0.043	0.79
Propulsion Battery	-6.7	0.60	-4.0
Payload	2.0	0.22	0.43
Payload servo	2.0	0.026	0.051
Elevator servo	20	0.026	0.53
Rudder servo	20	0.026	0.53
Ailerson Servo (x2)	2.1	0.053	0.11
ESC	-3.1	0.093	-0.28
Camera	-2.0	0.013	-0.026
Pixhawk	-5.1	0.068	-0.34

SUMMARY DATA	
Wingspan	39.1 in
Wing Area	264 inch²
Aspect Ratio	4.25
Empty Weight	3.05 lb
Battery(s) Capacity	2800mAh
Motor Model	F2203.5 Racingwhoop
Motor KV	2850KV
Propeller (x3)	APC 5x3.7
Servo	Corona DS843MG 63.9 oz-in torque

STABILITY	
Mean Chord	7.9 in
Stability Margin (Loaded CG)	10%
Stability Margin (Empty CG)	10%
Empty CG Behind Datum A	1.97 in
Fully Loaded CG Behind Datum A	1.97 in

TEAM NUMBER	206
TEAM NAME	U OF T AEROSPACE TEAM
SCHOOL NAME	UNIV OF TORONTO

UTAT UAS

UNLESS OTHERWISE SPECIFIED:
DIMENSIONS ARE IN INCHES
TOLERANCES:
LINEAR: X.X = ± 0.1; X.XX = ± 0.03; X.XXX = ± 0.01
ANGULAR: ± 0.3°

MATERIAL
Balsa, CFRP, Plywood

DO NOT SCALE DRAWING

TITLE:

UT-26 MAPLE

SIZE B | DWG. NO. #001-022626 | REV 1

SCALE: 1:9 | WEIGHT: 3.5 LB | SHEET 1 OF 1